

Лабораторная работа №6

Статическая маршрутизация VLAN

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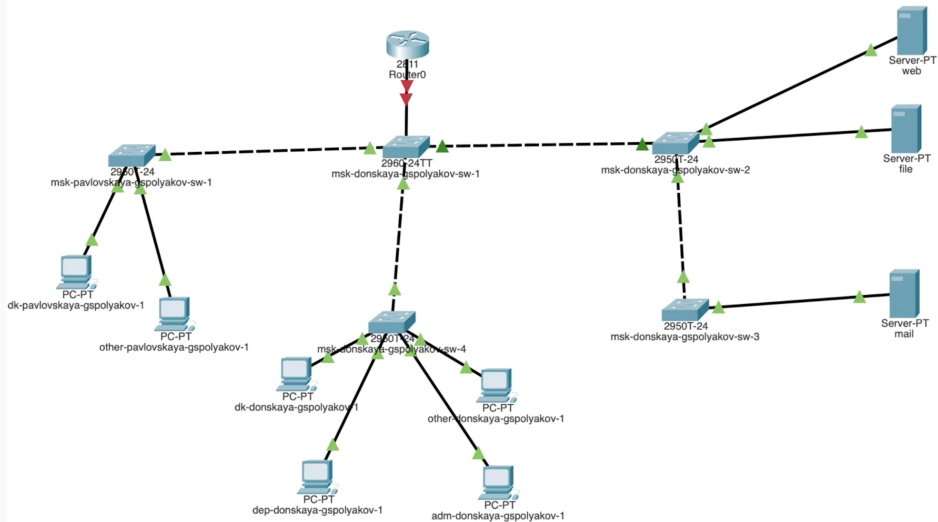
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Информация

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Выполнение лабораторной работы

Добавление маршрутизатора



Первоначальная настройка маршрутизатора

```
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname msk-donskaya-gspolyakov-gw-1
msk-donskaya-gspolyakov-gw-1(config)#lin
msk-donskaya-gspolyakov-gw-1(config)#line vty 0 4
msk-donskaya-gspolyakov-gw-1(config-line)#passw
msk-donskaya-gspolyakov-gw-1(config-line)#password cisco
msk-donskaya-gspolyakov-gw-1(config-line)#login
msk-donskaya-gspolyakov-gw-1(config-line)#ex
msk-donskaya-gspolyakov-gw-1(config-line)#exi
msk-donskaya-gspolyakov-gw-1(config-line)#exit
msk-donskaya-gspolyakov-gw-1(config)#enable secret cisco
msk-donskaya-gspolyakov-gw-1(config)#serv
msk-donskaya-gspolyakov-gw-1(config)#service pa
msk-donskaya-gspolyakov-gw-1(config)#service password-encryption
msk-donskaya-gspolyakov-gw-1(config)#username admin pr
msk-donskaya-gspolyakov-gw-1(config)#username admin privilege 1 secret cisco
msk-donskaya-gspolyakov-gw-1(config)#ip dom
msk-donskaya-gspolyakov-gw-1(config)#ip doma
msk-donskaya-gspolyakov-gw-1(config)#ip domain-name donskeya.rudn.ru
msk-donskaya-gspolyakov-gw-1(config)#ip domain-name donskeya.rudn.edu
msk-donskaya-gspolyakov-gw-1(config)#cryp
msk-donskaya-gspolyakov-gw-1(config)#crypto k
msk-donskaya-gspolyakov-gw-1(config)#crypto key g
msk-donskaya-gspolyakov-gw-1(config)#crypto key generate s
msk-donskaya-gspolyakov-gw-1(config)#crypto key generate r
msk-donskaya-gspolyakov-gw-1(config)#crypto key generate rsa
The name for the keys will be: msk-donskaya-gspolyakov-gw-1.donskeya.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]:
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

msk-donskaya-gspolyakov-gw-1(config)#lin
*Mar 1 0:1:16.442: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 0:1:16.443: %SSH-5-ENABLED: SSH 1.5 has been enabled
msk-donskaya-gspolyakov-gw-1(config)#line vty 0 4
msk-donskaya-gspolyakov-gw-1(config-line)#tra
msk-donskaya-gspolyakov-gw-1(config-line)#transport i
msk-donskaya-gspolyakov-gw-1(config-line)#transport input s
msk-donskaya-gspolyakov-gw-1(config-line)#transport input ssh
msk-donskaya-gspolyakov-gw-1(config-line)#exit
```

```
msk-donskaya-gspolyakov-sw-1(config)#interface f0/24
msk-donskaya-gspolyakov-sw-1(config)#interface f0/24
msk-donskaya-gspolyakov-sw-1(config-if)#acc
msk-donskaya-gspolyakov-sw-1(config-if)#in
msk-donskaya-gspolyakov-sw-1(config-if)#int
msk-donskaya-gspolyakov-sw-1(config-if)#swit
msk-donskaya-gspolyakov-sw-1(config-if)#switchport m
msk-donskaya-gspolyakov-sw-1(config-if)#switchport mode tru
msk-donskaya-gspolyakov-sw-1(config-if)#switchport mode trunk
```

Рис. 3: Настройка Trunk порта

Настройка VLAN интерфейсов

```
msk-donskaya-gspolyakov-gw-1(config)#interface f0/0
msk-donskaya-gspolyakov-gw-1(config-if)#inc
msk-donskaya-gspolyakov-gw-1(config-if)#inca
msk-donskaya-gspolyakov-gw-1(config-if)#enca
msk-donskaya-gspolyakov-gw-1(config-if)#no sh
msk-donskaya-gspolyakov-gw-1(config-if)#no shutdown

msk-donskaya-gspolyakov-gw-1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

msk-donskaya-gspolyakov-gw-1(config-if)#exit
msk-donskaya-gspolyakov-gw-1(config)#interface f0/0.2
msk-donskaya-gspolyakov-gw-1(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/0.2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.2, changed state to up

msk-donskaya-gspolyakov-gw-1(config-subif)#enc
msk-donskaya-gspolyakov-gw-1(config-subif)#encapsulation dot1Q 2
msk-donskaya-gspolyakov-gw-1(config-subif)#ip addr
msk-donskaya-gspolyakov-gw-1(config-subif)#ip address 10.128.1
                                     ^
% Invalid input detected at '^' marker.

msk-donskaya-gspolyakov-gw-1(config-subif)#ip address 10.128.1.1 255.255.255.0
msk-donskaya-gspolyakov-gw-1(config-subif)#des
msk-donskaya-gspolyakov-gw-1(config-subif)#description se
msk-donskaya-gspolyakov-gw-1(config-subif)#description mana
msk-donskaya-gspolyakov-gw-1(config-subif)#description management
msk-donskaya-gspolyakov-gw-1(config-subif)#exit
msk-donskaya-gspolyakov-gw-1(config)#interface f0/0.3
msk-donskaya-gspolyakov-gw-1(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/0.3, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.3, changed state to up

msk-donskaya-gspolyakov-gw-1(config-subif)#encapsulation dot1Q 3
msk-donskaya-gspolyakov-gw-1(config-subif)#ip address 10.128.0.1 255.255.255.0
msk-donskaya-gspolyakov-gw-1(config-subif)#description servers
msk-donskaya-gspolyakov-gw-1(config-subif)#exit
```


Настройка VLAN интерфейсов

```
msk-donskaya-gspolyakov-gw-1(config)#interface f0/0.101
msk-donskaya-gspolyakov-gw-1(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/0.101, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.101, changed state to up

msk-donskaya-gspolyakov-gw-1(config-subif)#encapsulation dot1Q 101
msk-donskaya-gspolyakov-gw-1(config-subif)#ip address 10.128.3.1 255.255.255.0
msk-donskaya-gspolyakov-gw-1(config-subif)#description dk
msk-donskaya-gspolyakov-gw-1(config-subif)#exit
msk-donskaya-gspolyakov-gw-1(config)#interface f0/0.102
msk-donskaya-gspolyakov-gw-1(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/0.102, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.102, changed state to up

msk-donskaya-gspolyakov-gw-1(config-subif)#encapsulation dot1Q 102
msk-donskaya-gspolyakov-gw-1(config-subif)#ip address 10.128.4.1 255.255.255.0
msk-donskaya-gspolyakov-gw-1(config-subif)#description departaments
msk-donskaya-gspolyakov-gw-1(config-subif)#exit
msk-donskaya-gspolyakov-gw-1(config)#encapsulation dot1Q 103
^
% Invalid input detected at '^' marker.

msk-donskaya-gspolyakov-gw-1(config)#interface f0/0.103
msk-donskaya-gspolyakov-gw-1(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/0.103, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.103, changed state to up
encapsulation dot1Q 103
msk-donskaya-gspolyakov-gw-1(config-subif)#ip address 10.128.5.1 255.255.255.0
msk-donskaya-gspolyakov-gw-1(config-subif)#descriptio
msk-donskaya-gspolyakov-gw-1(config-subif)#description adm
msk-donskaya-gspolyakov-gw-1(config-subif)#ex
msk-donskaya-gspolyakov-gw-1(config-subif)#exit
msk-donskaya-gspolyakov-gw-1(config)#encapsulation dot1Q 104
^
% Invalid input detected at '^' marker.

msk-donskaya-gspolyakov-gw-1(config)#encapsulation dot1Q 104
^
% Invalid input detected at '^' marker.
```

```
msk-donskaya-gspolyakov-gw-1(config)#interface f0/0.104
msk-donskaya-gspolyakov-gw-1(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/0.104, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.104, changed state to up
encapsulation dot1Q 104
msk-donskaya-gspolyakov-gw-1(config-subif)#ip address 10.128.6.1 255.255.255.0
msk-donskaya-gspolyakov-gw-1(config-subif)#description other
msk-donskaya-gspolyakov-gw-1(config-subif)#exit
msk-donskaya-gspolyakov-gw-1(config)#exit
```

Рис. 6: Настройка VLAN интерфейсов

```
msk-donskaya-gspolyakov-gw-1#ping 10.128.3.2  
  
Type escape sequence to abort.  
Sending 5, 100-byte ICMP Echos to 10.128.3.2, timeout is 2 seconds:  
.!!!!  
Success rate is 80 percent (4/5), round-trip min/avg/max = 0,5/23 ms  
  
msk-donskaya-gspolyakov-gw-1#
```

Рис. 7: Проверка доступности оконечных устройств

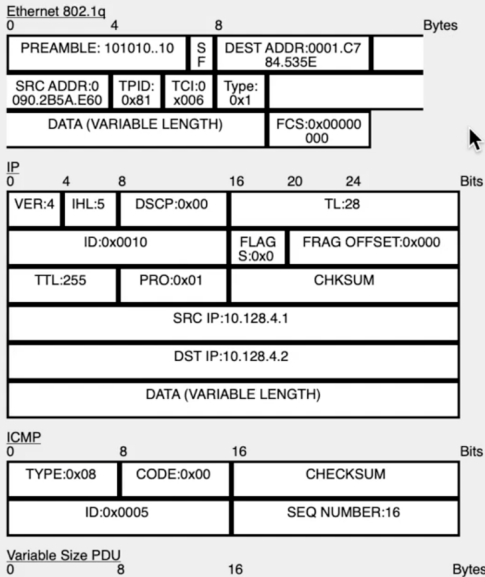
```
msk-donskaya-gspolyakov-gw-1#ping 10.128.4.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.128.4.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/6/25 ms

msk-donskaya-gspolyakov-gw-1#
```

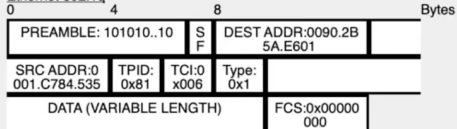
Рис. 8: Проверка доступности оконечных устройств

ICMP Packet

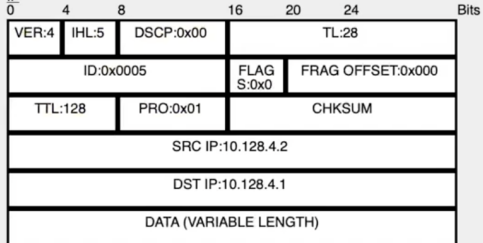


ICMP Пакет

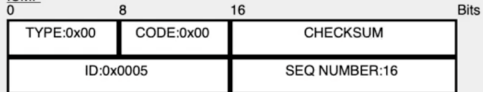
Ethernet 802.1q



IP



ICMP



Variable Size PDU

